DSA 3050A — Business Intelligence & Data Visualization

Mid-Semester Examination

Kenya Food Prices Analysis — Power BI Report

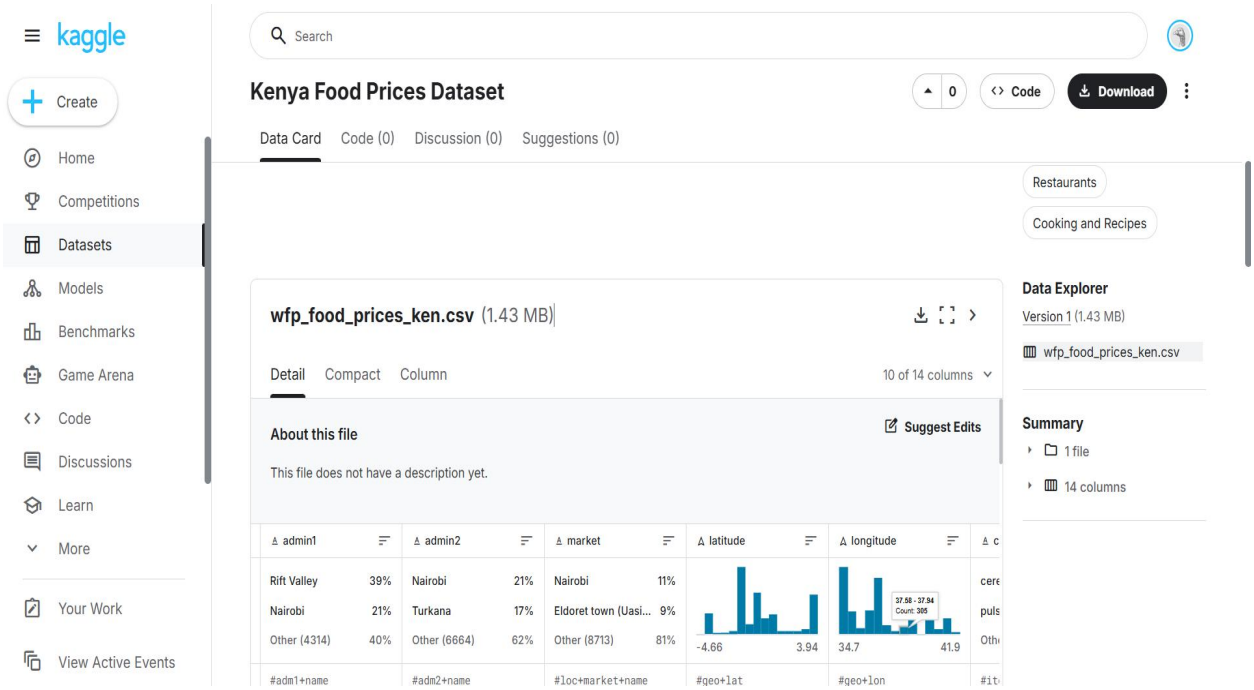
DATASET SOURCE EVIDENCE

This dataset was obtained from:

Kenya Food Prices Dataset — Kaggle
https://www.kaggle.com/datasets/usmanlovescode/kenya-food-prices-dataset

The dataset contains 10,000+ records and includes:

- Date
- Commodity
- Market
- Price
- Units
- Other descriptive attributes

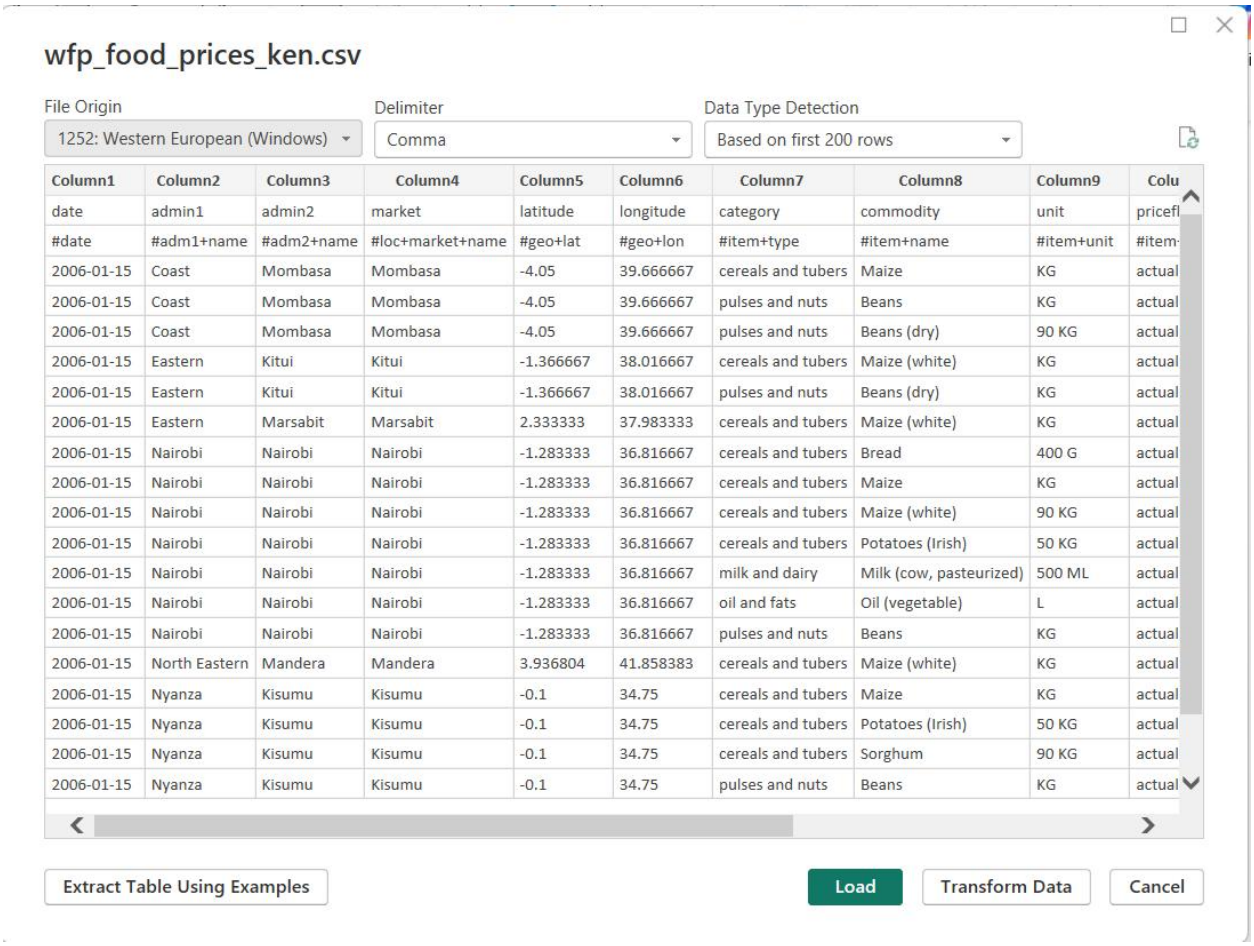


PART A — POWER QUERY

A1. Dataset Loading

The dataset was imported into Power BI Desktop and transformed.

Before transformation.



Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Colu
date	admin1	admin2	market	latitude	longitude	category	commodity	unit	pricefl
#date	#adm1+name	#adm2+name	#loc+market+name	#geo+lat	#geo+lon	#item+type	#item+name	#item+unit	#item-
2006-01-15	Coast	Mombasa	Mombasa	-4.05	39.666667	cereals and tubers	Maize	KG	actual
2006-01-15	Coast	Mombasa	Mombasa	-4.05	39.666667	pulses and nuts	Beans	KG	actual
2006-01-15	Coast	Mombasa	Mombasa	-4.05	39.666667	pulses and nuts	Beans (dry)	90 KG	actual
2006-01-15	Eastern	Kitui	Kitui	-1.366667	38.016667	cereals and tubers	Maize (white)	KG	actual
2006-01-15	Eastern	Kitui	Kitui	-1.366667	38.016667	pulses and nuts	Beans (dry)	KG	actual
2006-01-15	Eastern	Marsabit	Marsabit	2.333333	37.983333	cereals and tubers	Maize (white)	KG	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	cereals and tubers	Bread	400 G	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	cereals and tubers	Maize	KG	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	cereals and tubers	Maize (white)	90 KG	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	cereals and tubers	Potatoes (Irish)	50 KG	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	milk and dairy	Milk (cow, pasteurized)	500 ML	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	oil and fats	Oil (vegetable)	L	actual
2006-01-15	Nairobi	Nairobi	Nairobi	-1.283333	36.816667	pulses and nuts	Beans	KG	actual
2006-01-15	North Eastern	Mandera	Mandera	3.936804	41.858383	cereals and tubers	Maize (white)	KG	actual
2006-01-15	Nyanza	Kisumu	Kisumu	-0.1	34.75	cereals and tubers	Maize	KG	actual
2006-01-15	Nyanza	Kisumu	Kisumu	-0.1	34.75	cereals and tubers	Potatoes (Irish)	50 KG	actual
2006-01-15	Nyanza	Kisumu	Kisumu	-0.1	34.75	cereals and tubers	Sorghum	90 KG	actual
2006-01-15	Nyanza	Kisumu	Kisumu	-0.1	34.75	pulses and nuts	Beans	KG	actual

A2. Query Renaming

The main table was renamed to:

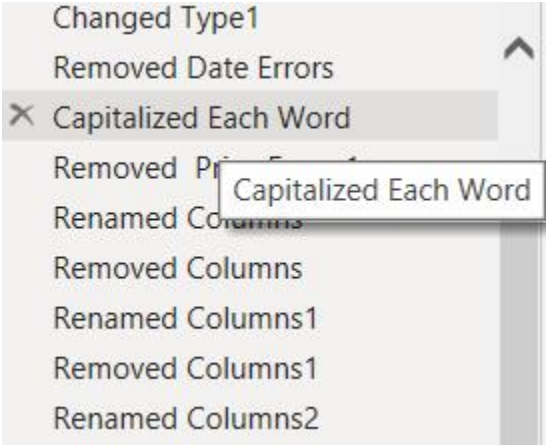
FactFoodPrices

A3. Data Cleaning & Transformation

Transformations were performed to enhance data quality.

1. Removal of Unnecessary Columns

Irrelevant columns were removed to improve model clarity and efficiency.

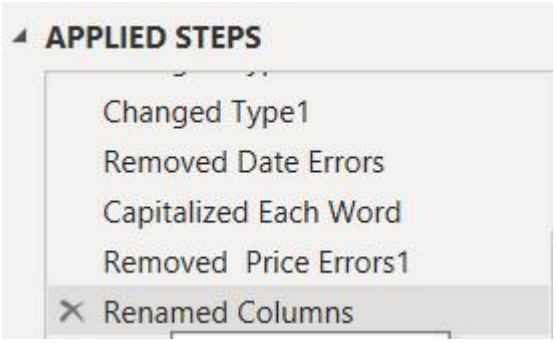


2. Removal of Duplicates

Duplicate rows were removed.

3. Handling Missing Values

Null values in date columns and other categorical fields such as Commodity and Market were removed because they were not significant to alter the data entirely.

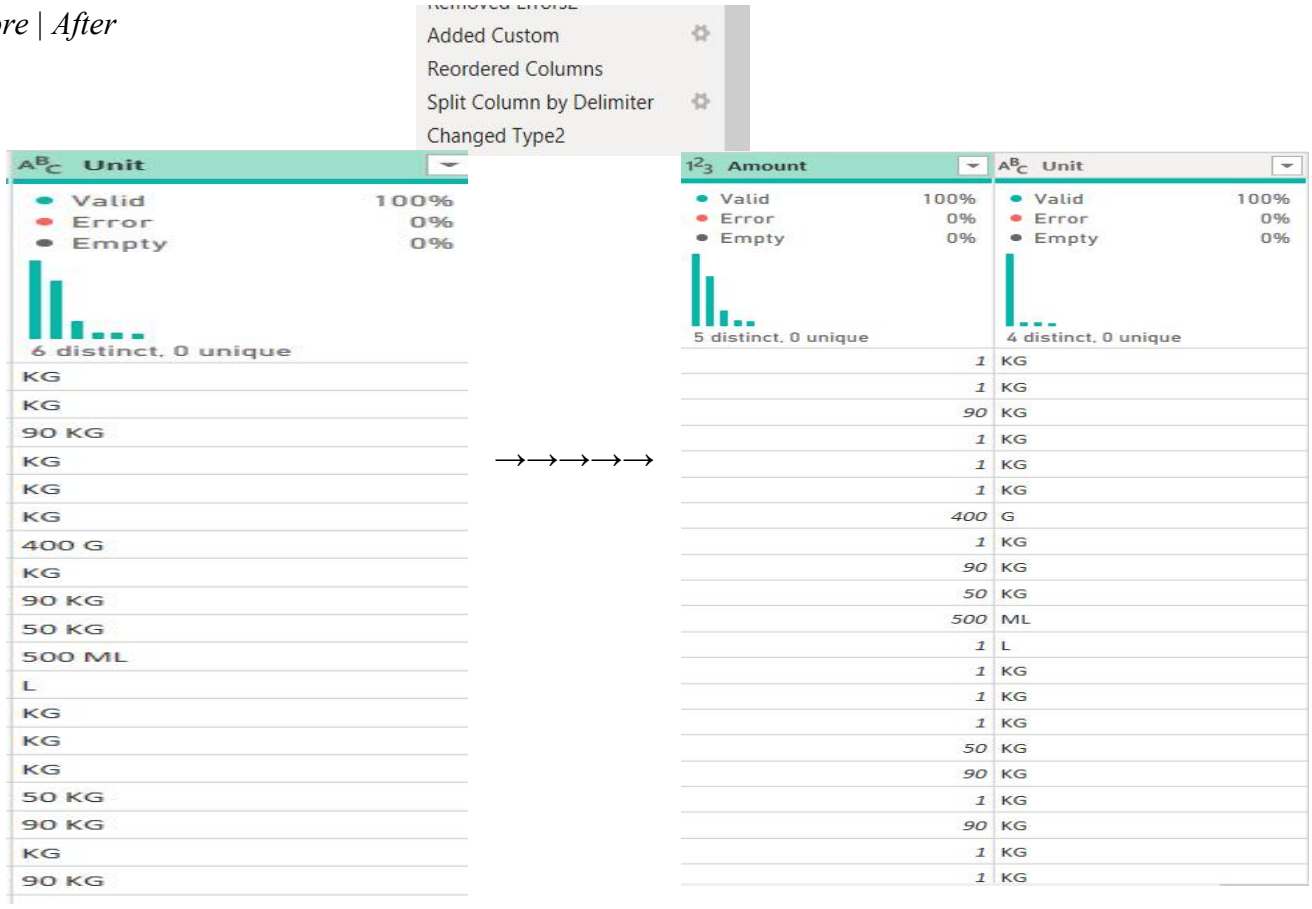


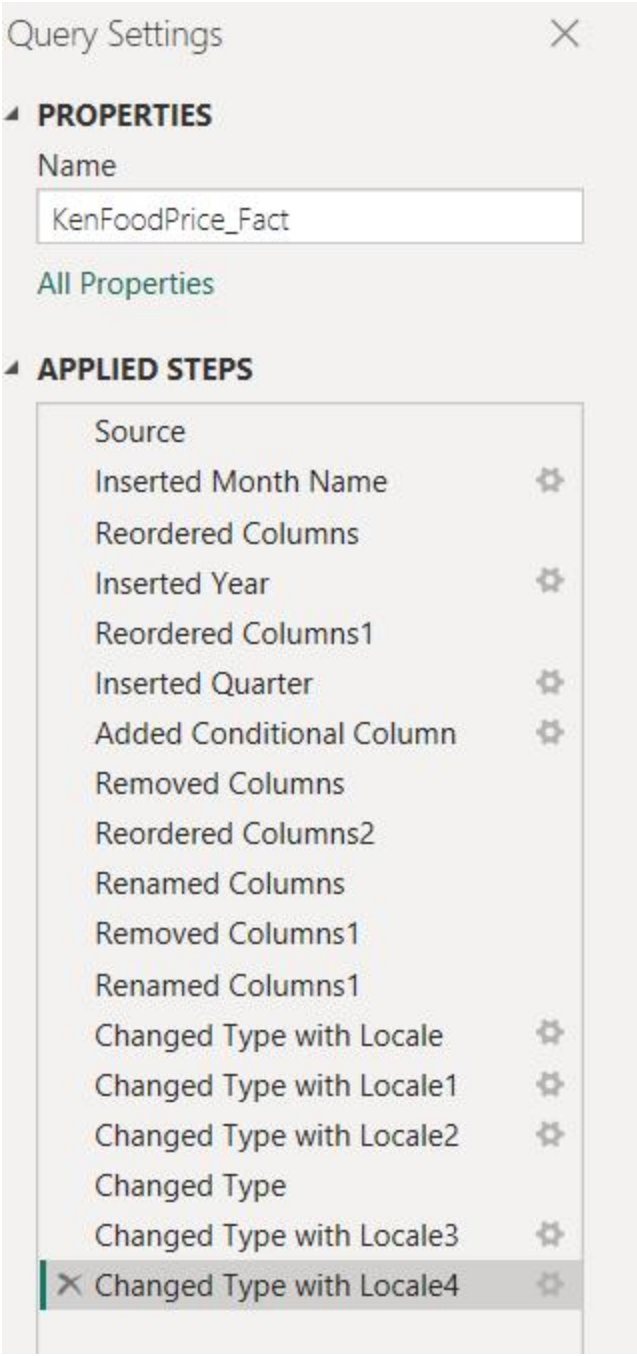
4. Data Type Corrections

The following corrections were made:

- Date - Date Type
- Price - Decimal Number
- Commodity - Text
- Market - Text
- Splitted Unit column by delimiter: from 90KG to |90|KG|

Before | After





5. Text Standardization

To ensure consistency:

- Trim applied
- Clean applied
- Capitalize Each Word applied

- Changed Type2
- Trimmed Text
- Cleaned Text
- Trimmed Text1
- Cleaned Text1
- Trimmed Text2
- Cleaned Text2
- Trimmed Text3
- Cleaned Text3

6. Date Attribute Extraction

Derived columns were created:

- Year
- Month
- Quarter

Before

	Date
	Valid 100%
	Error 0%
	Empty 0%
	48 distinct, 0 unique
1	15/01/2006
2	15/01/2006
3	15/01/2006
4	15/01/2006
5	15/01/2006
6	15/01/2006
7	15/01/2006
8	15/01/2006
9	15/01/2006
10	15/01/2006
11	15/01/2006
12	15/01/2006
13	15/01/2006
14	15/01/2006
15	15/01/2006
16	15/01/2006
17	15/01/2006

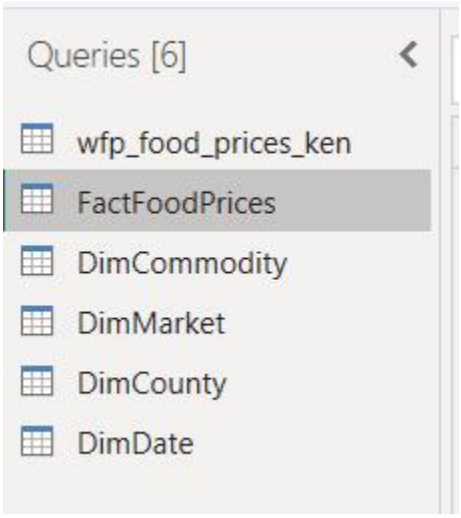
8.

After

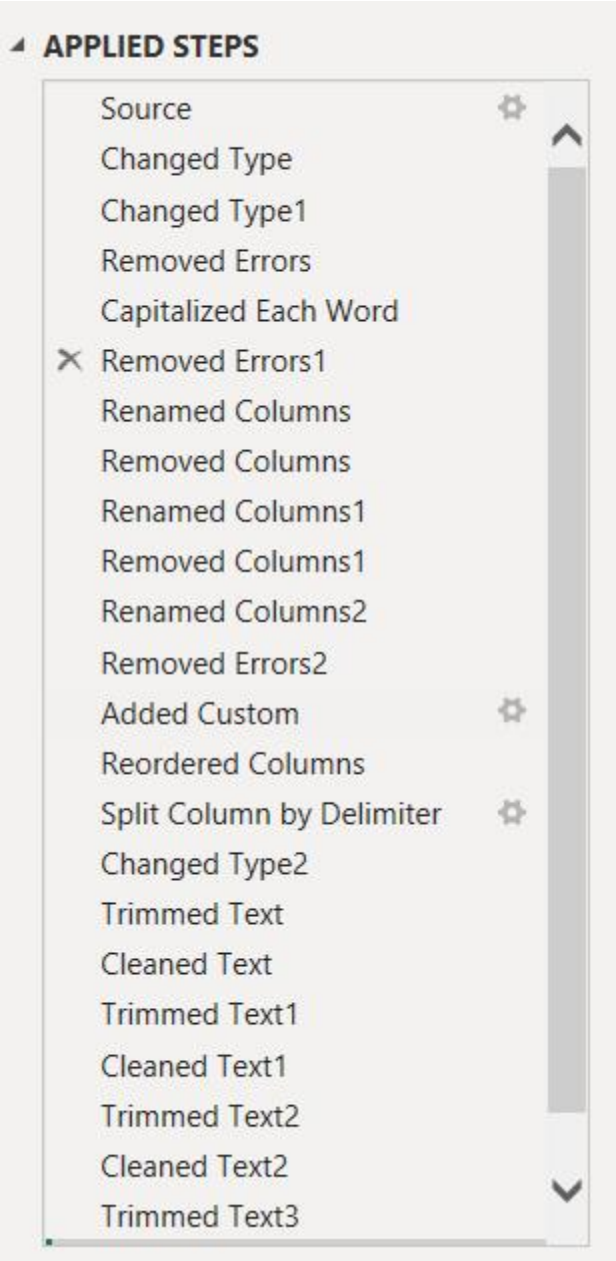
Quarter	Month	Year
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
	12 distinct, 0 unique	4 distinct, 0 unique
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006
Q1	January	2006

Query Dependency View

Table relationships within Power Query:



The Applied Steps pane shows more than 10 transformation steps……



PART B — DATA MODELLING

A star schema model was implemented.

B1. Creation of Dimension Tables

Using Reference from FactFoodPrices, the following dimension tables were created:

- DimCommodity
- DimMarket

- DimDate
- DimCounty

In each case, duplicates were removed and eac dimension table having a 1 to many * relationship with the fact table.

B2. Star Schema Structure

The model follows a star schema design where:

Fact Table:

- FactFoodPrices (contains transactional price data)

Dimension Tables:

- DimCommodity (descriptive commodity information)
- DimMarket (County level location information)
- DimDate (time intelligence attributes)
- DimCounty(counties in Kenya)

B3. Relationship Configuration

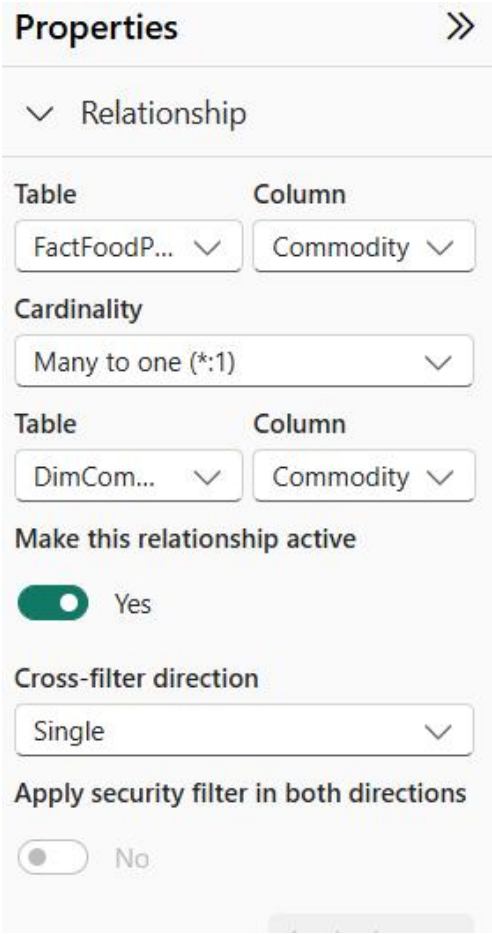
The following relationships were created:

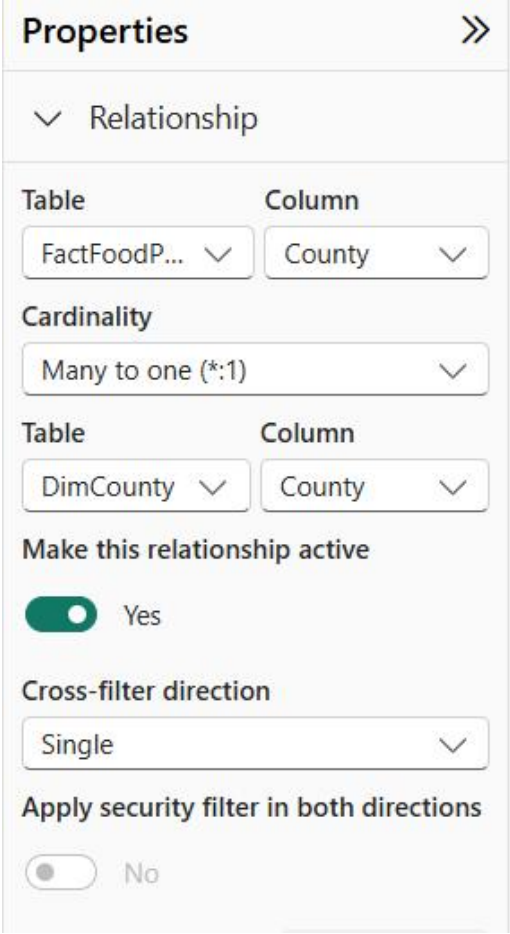
1. DimCommodity - FactFoodPrices
2. DimMarket - FactFoodPrices
3. DimDate - FactFoodPrices
4. DimCounty - FactFoodPrices

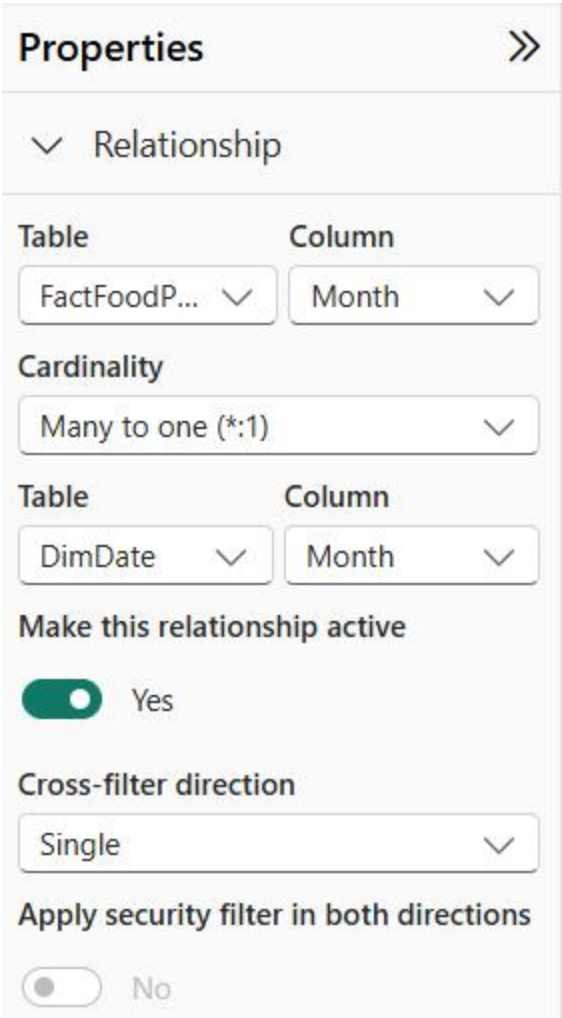
Each configured as:

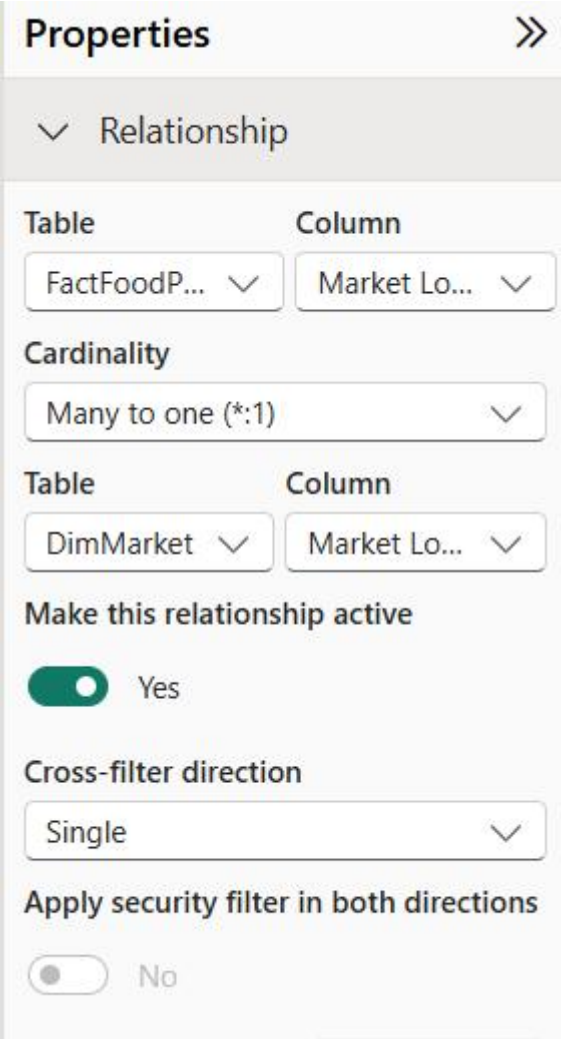
- One-to-Many
- Single cross-filter direction

- Active relationship



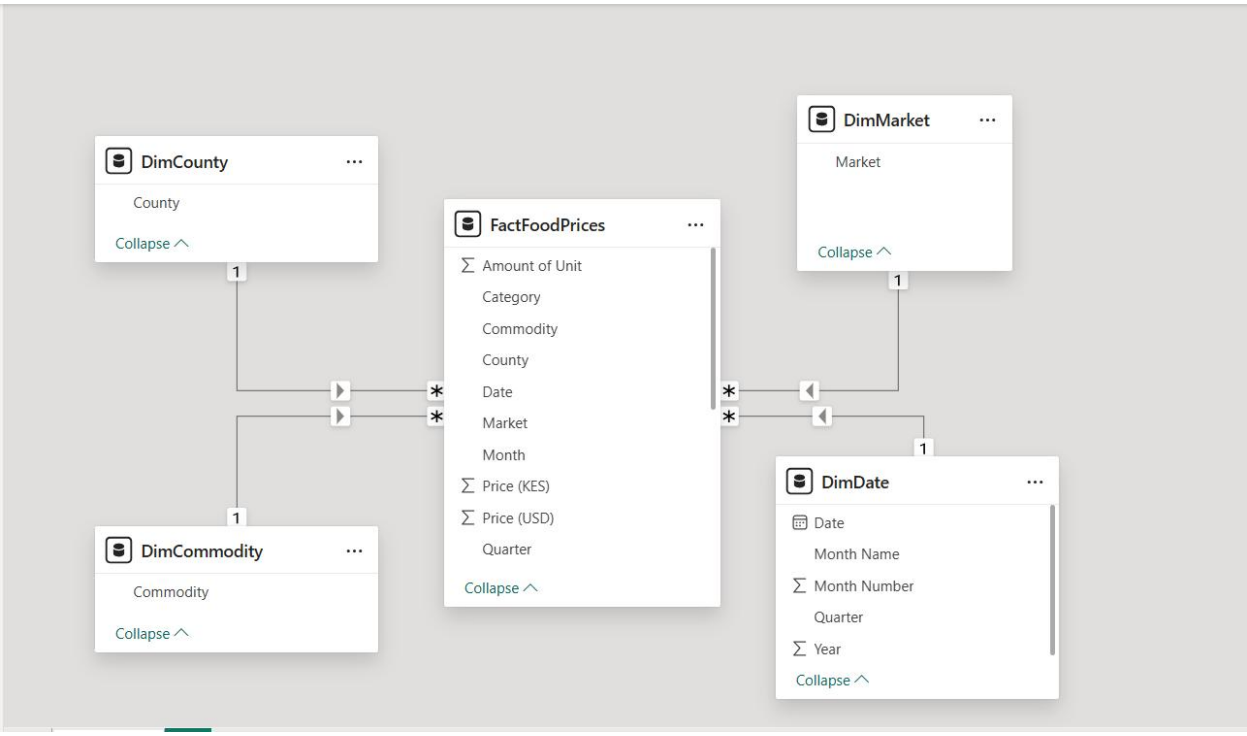






B4. Model Organization

- Fact table positioned centrally
- Dimensions arranged around fact
- Technical columns hidden
- No many-to-many relationships



PART C — DASHBOARD DEVELOPMENT

C1. KPI Summary

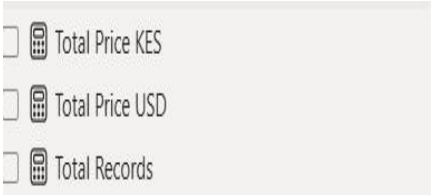
The following DAX measures were created:

Total Records:

Maximum Price:

Minimum Price:

Total Price(USD/KES):



KPI Summary screenshot

Overall Market Performance

- ***Total Price (KES): 20.35M***
  The dataset represents a significant market value across commodities and counties.

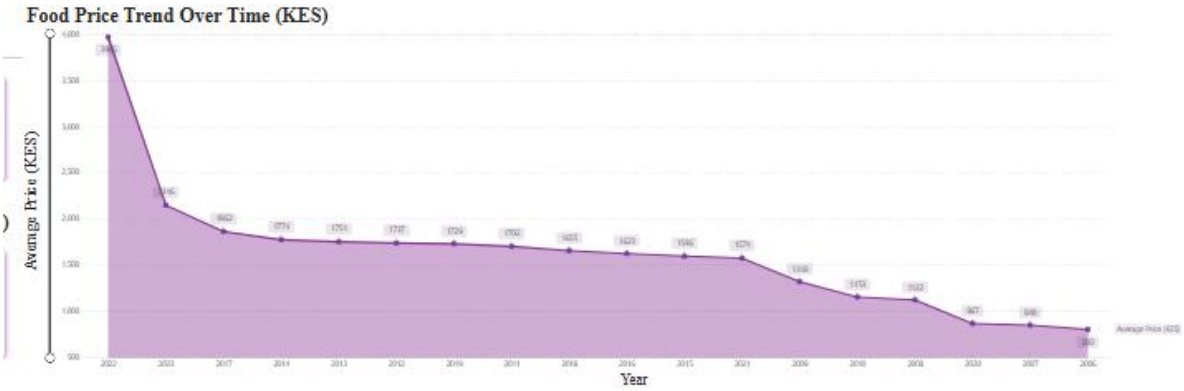
- ***Total Records: 11K***
  Strong dataset size, indicating reliable trend analysis.

- ***Minimum Price: 5 KES***
  Large price variability exists across commodities or regions

Price Spread

- *Price Spread = 18K*
  Large difference between min and max pricing.
  Market has strong variability.
  Potential seasonal or commodity-driven fluctuations

C2. Trend Analysis



Shows food price trends over time.

Price Trend Over Time

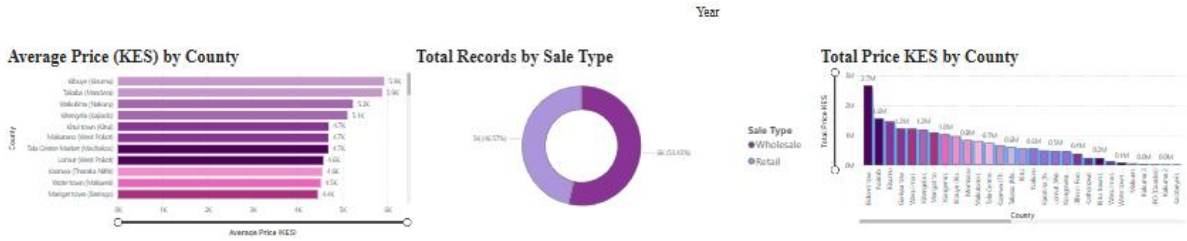
- *Prices show a **strong decline trend over the years**.*
- *There was an early peak period followed by gradual stabilization.*
- *Recent years indicate **price compression**, suggesting:*
    - *Increased supply*
    - *Market stabilization*
    - *Policy intervention*
    - *Or commodity mix shift*

The long-term trend suggests weakening commodity price levels

C3. Comparative Analysis

Bar Chart:

- Axis: Commodity
- Values: Average Price
- By County
- Total Records
- By SaleType



County Comparison

- *Price levels vary significantly by county.*
- *Some counties consistently record higher total prices.*
- *This suggests:*
    - *Regional supply-demand imbalance*
    - *Transport/logistics cost differences*
    - *Production concentration zones*

Geographic price disparity is evident.

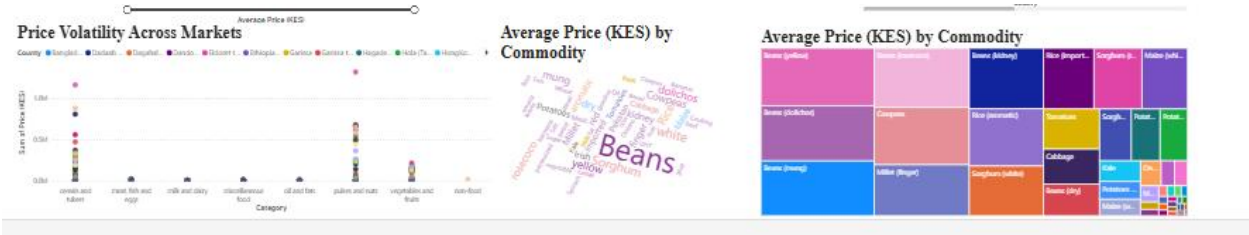
Commodity & Category Insights

- *Certain commodities dominate total price contribution.*
- *Category filters show clear clustering of price levels.*
- *Indicates:*
    - *Commodity-specific volatility*
    - *Certain agricultural products drive overall value*

C4. Distribution Analysis

Price volatility across markets

Average price by commodity



Sale Type Distribution

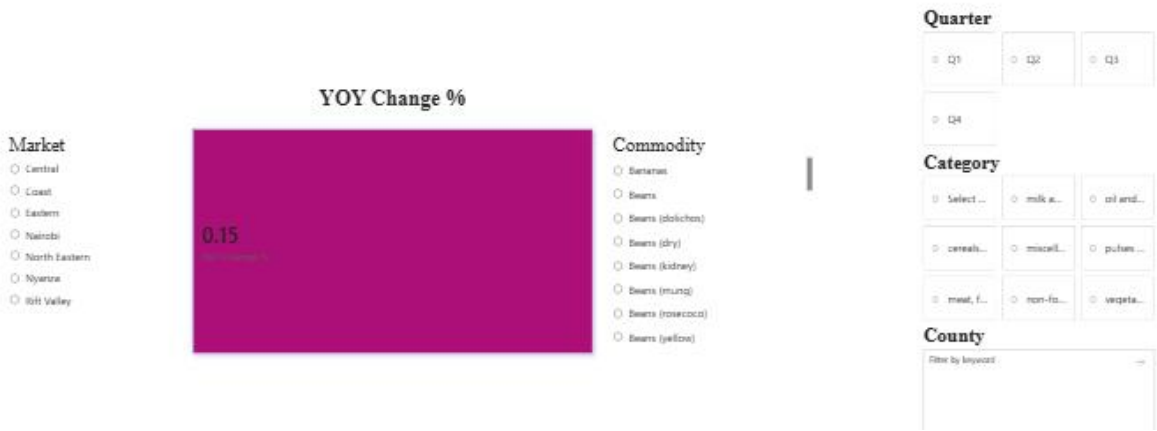
- *Retail and Wholesale show clear distribution differences.*
- *One category dominates record count.*

- *Suggests:*
    - *Market structure is skewed toward one trade channel.*
    - *Price behavior differ by sale type.*

C5. Interactive Slicers

Slicers added:

- Commodity
- Market
- Year
- Quarter
- Category

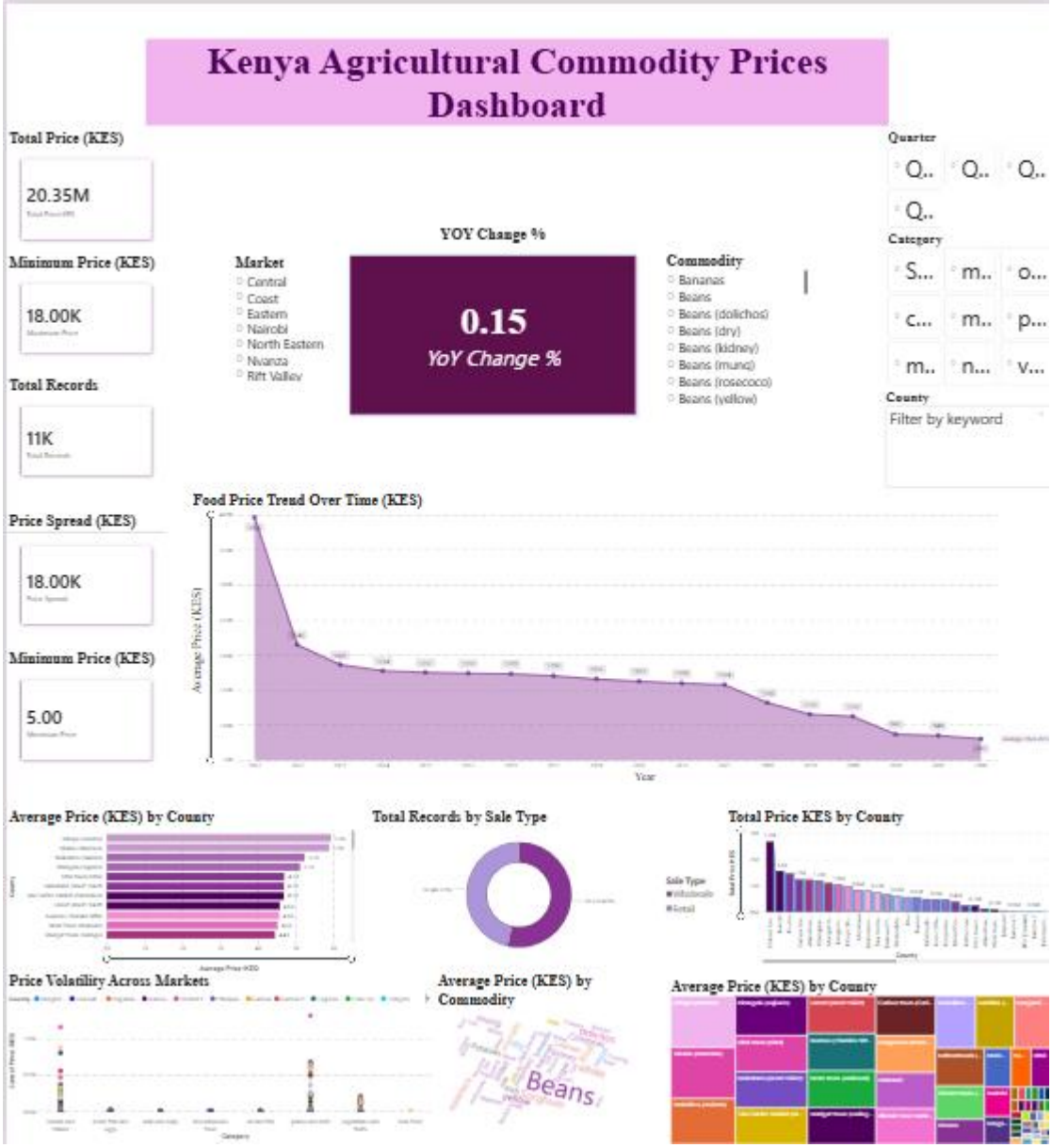


Year-Over-Year (YoY) Change

- *YoY Change % ≈* ***+15% (0.15)***
  Despite long-term decline, the most recent year shows recovery growth.
  Indicates short-term upward trajectory of price stabilization.

PART D — PUBLISHING & DEPLOYMENT (5 MARKS)

The report was published to Power BI Service.



Power BI dashboard view.

Power BI Public Dashboard Link:
[Martin's Midsem Dashboard View link]

Conclusion

From the dashboard, it can be inferred that Kenya's agricultural commodity market has experienced noticeable changes over time. Although prices generally show a long-term decline trend, the recent Year-over-Year growth of about 15% suggests that the market may be starting to recover.

There are clear differences in pricing across counties, which indicates that location plays a big role in agricultural pricing. This could be due to factors like supply levels, transportation costs, or production capacity in certain regions.

The price spread is also quite wide, meaning there is significant variation between the lowest and highest prices. This shows that some commodities or markets are much more expensive than others.

Additionally, one sale type seems to dominate the market, suggesting that the structure of trade is not evenly distributed between retail and wholesale channels.

Overall, the agricultural market appears to be stabilizing after a period of decline, but regional disparities and commodity differences still strongly influence pricing patterns.

Recommendations

- The government and market regulators should monitor recent price increases closely to ensure that the upward trend reflects healthy market recovery rather than inflationary pressure.
- Counties with significantly higher prices should be further analyzed to understand whether transport costs, supply shortages, or market inefficiencies are driving the differences.
- Efforts should be made to reduce price disparities across regions by improving distribution networks and market access.
- Since certain commodities dominate total market value, stakeholders should prioritize stabilizing these key products to maintain overall market balance.
- A balanced development of both retail and wholesale channels is recommended to ensure fair competition and better price regulation.
- Continuous monitoring of price trends using data-driven dashboards should be encouraged to support timely decision-making.